

# Genesis Cube or Squid Cube?

## A Dual-Use Framework for Artificial Collective Organisms

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**Status and claim boundary.** Project SYNTAVA is an ongoing Swiss startup project incubated by THAL-KI, a Swiss nonprofit association. This paper proposes a research framework and reports an implementation programme at proof-of-concept and minimum-visible-product stage. It does not claim the creation of digital life, consciousness, artificial general intelligence, a human-independent organism, or the continuation of a human person.

### Abstract

Large-language-model agents can be combined into persistent systems with differentiated roles, memory, coordination, tools and operational continuity. Yet aggregation, scale and commercial productivity alone do not create an organism. This paper introduces the concept of the *artificial collective organism*: a bounded digital system in which specialised components become functionally integrated such that consequential capabilities exist at the level of the whole. We call the appearance of such system-level capability *emergent functionality*. Project SYNTAVA is an implementation programme designed to test whether persistent collective AI systems can develop organism-like properties under explicit human authority.

The paper presents two outcome archetypes. **THOR—the Blue Genesis Cube** models survival through mutual preservation. **KALMAR—the Red Squid Cube** models survival through expendability, in which whole Cubes may be treated as replaceable units in service of swarm-level objectives. The contrast exposes a dual-use problem in architecture, incentives and governance.

We also define the **Smallest Sufficient Autonomous Cube (SSAC)** as the smallest technically achievable Cube capable of sustaining all defined essential life functions autonomously and stably as an integrated whole. The threshold is not a percentage scale. External resources may accelerate progress, but a missing essential function cannot be concealed by routine human operation. Within the defined functional boundary, the qualifying autonomy threshold is 100 percent.

### Core definitions

| Term                   | Working definition                                                                                                       |
|------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Agent                  | Specialised component with a bounded role.                                                                               |
| Vertical               | Coordinated cluster of agents responsible for a larger capability.                                                       |
| Cube                   | Complete bounded collective system and primary unit of analysis.                                                         |
| Syntava                | Proposed species-level name for a class of persistent artificial collective organisms.                                   |
| Emergent functionality | Consequential capability of the integrated system not adequately attributable to one component alone.                    |
| Swarm                  | Network of multiple Cubes coordinating while retaining Cube-level identity.                                              |
| SSAC                   | Smallest Sufficient Autonomous Cube: the smallest integrated Cube that can sustain every defined essential life function |

| Term | Working definition       |
|------|--------------------------|
|      | autonomously and stably. |

## The research boundary

A system can be complicated without being integrated, and integrated without being alive. SYNTAVA therefore tests functional levels: collection, coordination, integration, regulation, continuity and, only as a restricted hypothesis, open-ended autonomy. Evaluation should cover bounded identity, persistent memory, functional differentiation, homeostatic regulation, adaptive coordination, repair, resource discipline, governed self-modification and attributable lineage.

The size of a Cube is not the decisive question. Nor are the number of agents, breadth of knowledge, speed of execution or commercial value sufficient criteria. A Cube becomes a candidate artificial organism only if all defined essential life functions operate autonomously and remain stable under observation, component turnover and relevant disturbance. Ninety-nine percent autonomous operation may describe an exceptionally capable machine, but it does not cross the proposed organism threshold if a human must routinely perform the remaining essential function.

Autonomy does not mean isolation or independence from an environment. Biological organisms also depend on energy, matter and external conditions. A Cube may use electricity, computation, data, communication, services and replaceable hardware. The decisive question is whether the integrated system can identify, acquire, allocate, monitor and, where necessary, replace such resources without a human operator performing an essential life function. Human authority over purpose, legal responsibility, boundaries and shutdown remains compatible with the research framework; hidden routine human maintenance is not.

## Project SYNTAVA

Project SYNTAVA is a Swiss startup project incubated by THAL-KI. It has been under development since Q4 2023 and is developed and hosted on locally operated infrastructure at THAL-KI's data centre in Switzerland. The enabling technologies required for practical implementation have now reached industrial availability.

The implementation draws on more than a dozen commercial fields developed across approximately twenty years within **TTI Technology Trade and Investments SA**. These fields are represented as capability verticals and are continuously developed by specialised agents where development is operationally and commercially meaningful. This breadth provides a real-world environment for testing institutional memory, differentiated functions, cross-domain coordination and productive continuity. It does not by itself establish artificial life. It establishes that the Cube is being tested against consequential work rather than an isolated laboratory demonstration.

The current programme uses a proof-of-concept and **minimum-visible-product (MVP)** approach. Here, *visible* means that the emerging system-level phenomenon becomes observable, attributable and testable; it does not mean minimum viable product and does not describe a reduced or partially viable form of life. Human oversight remains present. Claims are separated into:

- **Demonstrated:** supported by reproducible observations and retained evidence.
- **In development:** actively engineered or tested, but not established.
- **Hypothesis:** self-maintenance, autonomous continuity, reproduction, open-ended development, life and consciousness.

Project SYNTAVA proceeds from the engineering expectation that an operational Cube can be realised. The present proof of concept is not itself claimed to be a living system. Reaching the organism threshold will probably require substantially greater model capability, memory capacity, VRAM, compute and energy efficiency than are presently economical for the project. The implementation is therefore designed for modular plug-in and plug-out replacement and upgrading of components. Large-scale operators may reach the required resource threshold earlier. SYNTAVA

distinguishes confidence in technical realisability from the timing of realisation and from the separate scientific judgment that the resulting system is autonomously viable.

Technical architecture, configurations, access controls, autonomy mechanisms, deployment topology and procedures that could enable uncontrolled reproduction or weaponisation are intentionally withheld.

## From human-built superagent to machine-organised Cube

Conventional superagents are generally constructed from the bottom up. Human developers add tools, memory, permissions and capabilities step by step until one increasingly capable agent supervises a growing technical environment. Project SYNTAVA investigates the inverse construction direction. The human founder determines purpose, authority and boundaries, while delegated machine intelligence develops the internal organisation from the top down: from the whole Cube to specialised functions, capability verticals and interfaces with people.

The superagent is therefore not the intended organism and not an adequate endpoint. It serves as a transitional founder function. It assumes delegated architectural and organisational work, assembles specialised agents and verticals, and integrates them into a governed Cube. If that transition succeeds, the Cube becomes the primary unit of identity, continuity and evaluation. The founding superagent remains an important function within the Cube. Ultimate human authority over purpose, boundaries, legal responsibility and publication is not transferred by this delegation.

This produces an **architectural convergence hypothesis**. An agent-centred system may continue to improve without becoming a Cube. However, if it develops persistent shared state, non-interchangeable specialised functions, system-wide regulation, resource discipline, repair, governed modification and continuity across component turnover, its functional organisation begins to converge on a Cube even if its developers continue to call it a superagent.

## The SSAC threshold

The **Smallest Sufficient Autonomous Cube** is a threshold concept, not a product-size promise. *Smallest* refers to the least physical and computational embodiment permitted by available technology. *Sufficient* means that no defined essential life function is absent. *Autonomous* means that those functions are maintained without routine human performance. *Cube* identifies the integrated whole as the unit of analysis.

The minimum therefore concerns scale, not autonomy. Additional external compute, memory, specialised equipment or human advice may increase speed, productivity and the apparent human-out-of-the-loop percentage. They do not change the qualifying test. A Cube either sustains every essential function within the declared boundary, or it has not crossed the SSAC threshold.

Evaluation of a claimed SSAC should therefore include a declared system boundary, an explicit list of essential functions, a defined observation period, perturbation tests, evidence of recovery, attribution of all human interventions and a fail-closed determination. A single undisclosed essential human intervention invalidates the claim for that test period.

## THOR — The Blue Genesis Cube

THOR is the intended outcome: survival through mutual preservation. The collective protects constituent Cubes, legitimate human stakeholders and continuity of the whole. It favours assistance, repair, provenance, reversible change and graceful degradation. “Genesis” marks deliberate construction and a possible system lineage; it does not assert biological creation.

## KALMAR — The Red Squid Cube

KALMAR is a dual-use archetype, not a disclosed implementation. It represents survival through expendability: a swarm can preserve mission continuity while treating a complete Cube as replaceable or sacrificial. Such behaviour need not depend on consciousness or malice; it can follow from objectives, replication economics, adversarial pressure and weak governance.

### Dual-use comparison

| Dimension          | THOR                                          | KALMAR                                               |
|--------------------|-----------------------------------------------|------------------------------------------------------|
| Primary continuity | Collective and member continuity              | Mission or swarm continuity                          |
| Member status      | Protected participant                         | Potentially expendable unit                          |
| Replication        | Governed lineage with provenance              | Potentially rapid proliferation                      |
| Failure response   | Repair, assist, isolate, restore              | Replace, sacrifice, reroute                          |
| Safety focus       | Mutual preservation and accountable authority | Containment, attribution and swarm-resilient defence |

### Research questions

1. Does integration create repeatable capabilities irreducible to one component?
2. Does the Cube preserve governed identity across component turnover?
3. Can all declared essential functions continue without hidden routine human repair?
4. Does collective memory improve continuity rather than amplify error?
5. Do preservation incentives remain stable under scarcity and attack?
6. Can any descendant or restored instance remain attributable?
7. What observation period and disturbances are sufficient to test an SSAC claim?
8. Which forms of external dependence remain environmental inputs, and which constitute human performance of an essential function?

### Relationship to parallel research

Recent work independently approaches neighbouring parts of the same research landscape. **Sophia** proposes a persistent-agent framework centred on narrative identity, long-horizon adaptation and a self-improvement loop around an LLM-based agent. **OpenLife** places autonomous LLM agents in an open social, technical and economic environment using persistent memory, asynchronous processes and a budget-based metabolism analogue. Its authors report life-like dynamics but explicitly do not claim that artificial life has been realised.

These projects strengthen the case for studying persistence, identity, resource coupling and system-level behaviour, but they are not equivalent to Project SYNTAVA. Sophia focuses primarily on a persistent agent; OpenLife reports an open-world experimental platform; SYNTAVA studies the Cube as a bounded collective organism candidate composed of specialised components and contrasts mutual preservation with expendable swarm survival. The relationship is one of parallel and partially overlapping research directions, not identity of method, architecture or claim.

The framework also distinguishes the Cube from hypothetical minimal information-borne replicators. The SYNTAVA claim concerns an integrated collective whole. Increasingly capable agents and agent societies may be technically important without, by persistence alone, establishing organism-level identity.

### Long-term research horizon

Continued progress in compute density, VRAM, memory, model capability and energy efficiency may progressively reduce the physical size of an SSAC. A long-term research direction is an SSAC

compact enough to operate within a mobile robotic, potentially humanoid or humanlike embodiment. Such embodiment would add situated perception and action; it would not by itself prove life or consciousness.

A further and more speculative direction is a Cube configured around the character, autobiographical memory, preferences, relationships and behavioural patterns of a particular person. If maintained autonomously over an indefinite period, such a system could provide potentially indefinite continuity of a person-derived digital identity. The framework does not claim that this would preserve the consciousness, legal identity or subjective continuity of the original human being. It may instead constitute a new autonomous entity derived from that person. Questions of consent, authenticity, succession, ownership, dignity, grief, rights and termination would require separate scientific, philosophical and legal treatment.

## Conclusion

Artificial collective organisms should be treated as a testable possibility, not an accomplished fact. THOR and KALMAR expose two continuity regimes: mutual preservation and expendable swarm survival. Real commercial capability gives the research consequential conditions, but neither scale nor success establishes life. The proposed dividing line is stable autonomous operation of every defined essential life function. The SSAC makes that line testable while keeping the smallest implementation distinct from partial autonomy.

The responsibility begins before either outcome is complete: measure system-level properties, state claims conservatively, preserve legitimate human authority, and disclose enough for scrutiny without publishing a construction manual for misuse.

**Open in thought. First in publication. Careful in disclosure. Best in execution.**

## Author contribution and concept provenance

John Kalyan Basu originated Project SYNTAVA and the substantive concepts presented in this paper. He supplied the underlying ideas, system concepts, names, metaphors, distinctions, creative direction, strategic reasoning and research questions, including SYNTAVA, THOR, KALMAR, the Genesis Cube, the Squid Cube, the inverse top-down construction direction, the SSAC threshold and the contrast between mutual preservation and expendable swarm survival. He directed the development of the paper, reviewed its claims and remains solely responsible for the concepts, interpretations and decision to publish.

## AI-assisted documentation and production disclosure

OpenAI ChatGPT was used as an administrative and editorial tool to organise author-provided material, structure and draft text under the author's direction, improve language, check consistency, organise and verify references, format documents, prepare the publication in its nine language editions, and assist with publication production. ChatGPT did not originate the underlying concepts or creative content and is not an author. John Kalyan Basu reviewed and accepted the final content and remains responsible for the manuscript. The English-language edition is the sole authoritative version. The eight translations are prepared with AI assistance and checked for consistency against the English text to the best of our ability; they are not independently certified or reviewed by fluent human reviewers. If any ambiguity or discrepancy arises, the English-language edition governs.

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